

Oscillatory State Persistence and Stable Video Infinity

Error Memory, Corrective Dynamics, and a Testable Neural Analogy

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July 2026

Abstract

A system can make each local step look plausible while progressively losing the identity of the process it is supposed to preserve. In long video generation, a face changes, a room drifts, an object's geometry mutates, or the storyline ceases to follow its prompt. Stable Video Infinity (SVI) treats this as a mismatch between training on clean histories and testing on histories already contaminated by the model's own errors. Its Error-Recycling Fine-Tuning exposes the model to historical errors, estimates corrective differences, stores them in Error Replay Memory, and learns a velocity from corrupted latent states toward clean targets.

This paper asks what general principle can be extracted from that engineering mechanism and whether a related problem exists in biological cognition. It first states a paper-level control model in which present mismatch and a decaying history of mismatch jointly guide correction. A preregistered bounded surrogate with 200 fixed seeds, 400 steps, and three disturbance regimes found that replay-informed correction reduced mean trajectory error by 5.80-6.30% relative to instantaneous correction and by 89.95-94.26% relative to open-loop propagation. A disclosed high-gain stress condition failed, showing that persistent correction can destabilize a system when memory is coupled too strongly.

The paper then develops a constrained Self Aware Networks (SAN) hypothesis. Slow recurrent context can provide a temporal reference against which faster departures become informative. A receiver-relative departure that changes the next update is called a Phase Wave Differential (PWD). Repeated retrieval, assessment, correction, and confirmation is called mulling. Predictive coding, oscillatory communication, cross-frequency coupling, and corticothalamic gating provide established components, but not a completed biological implementation of the proposed operator. The resulting contribution is a testable bridge: long-horizon intelligence requires not only memory of state, but memory of how state has been departing from its intended trajectory.

Keywords: long-horizon drift; Stable Video Infinity; error recycling; state persistence; predictive coding; neural oscillations; corticothalamic feedback; Phase Wave Differential; Self Aware Networks

Scope and evidence boundary

selfawarenetworks.com Program role: Paper 03

1. The ordinary problem: a sequence can look right while becoming wrong

Imagine watching a generated video of a person walking through a house. Every new frame is locally convincing. Yet after several minutes the person's face has changed, the room no longer has the same dimensions, an object has moved without cause, and the story has lost its original instruction. Nothing required a single catastrophic error. Small deviations accumulated until the sequence was no longer the same sequence.

The same structural problem appears in other long-running systems. A planner can satisfy each immediate subtask while departing from its mission. A conversation can answer each sentence while forgetting the

user’s governing constraint. A biological organism can receive familiar sensory fragments while failing to maintain the body, object, goal, and temporal context that make those fragments part of one continuing situation.

The problem is therefore not merely storage. It is trajectory preservation. The system must:

1. retain enough context to define what continuity means;
2. detect departures from that context;
3. preserve information about recurrent departures rather than discarding it;
4. produce a corrective update;
5. check whether correction survives the next cycle.

This paper uses **semantic decoherence** as a local name for progressive loss of trajectory identity. The term does not assert quantum decoherence in video models or brains. It describes a practical condition in which an inferred or generated state moves away from the task, object, body, story, or context it is supposed to preserve.

2. An engineering solution explained before it is named

Training a long-video model only on clean sequences creates a hidden weakness. During actual autoregressive use, the model conditions on its own prior outputs. Those outputs already contain errors. The model is therefore asked at test time to recover from states it did not learn to repair during training.

A direct response is to make prior failure part of the training material. Let the model encounter realistic historical errors, estimate how the corrupted state differs from a clean target, keep a reusable collection of those differences, and train the direction that moves the corrupted state toward recovery. The error is no longer disposable residue. It becomes information about how the system fails and how it can correct itself.

Stable Video Infinity names this procedure **Error-Recycling Fine-Tuning (ERFT)**. In the arXiv-v1 mechanism, historical-error injection simulates accumulated degradation; one-step bidirectional integration curates error estimates; Error Replay Memory banks errors across discretized timesteps; and an error-recycled velocity trains movement from the error-injected state toward a clean latent target [1]. The authors report evaluations on three long-video benchmarks and describe the intervention as a training-time method without added inference cost [1].

SVI mechanism	Functional role in the engineering system
Training-test hypothesis gap	Identifies that clean training histories differ from error-bearing autoregressive histories.
Historical-error injection	Exposes the model to the kind of accumulated degradation it will face during use.
Bidirectional error curation	Estimates corrective differences using a model-specific one-step integration procedure.
Error Replay Memory	Retains and resamples mismatch examples across discretized timesteps.
Error-recycled velocity	Learns a direction from corrupted latent state toward clean target.

The important innovation is distributional. SVI does more than append a memory module to a generator. It changes what the model learns from: the model’s own characteristic error trajectories become supervised material.

3. A paper-level model of persistent correction

The following equations formalize the general principle tested in this paper. They are not presented as SVI’s published equations and do not reproduce the SVI model. They define a transparent control abstraction for asking whether a history-bearing mismatch state can improve persistence.

Let x_t be the current system state and r_t the reference state at step t . The instantaneous mismatch is

$$e_t = x_t - r_t. \quad (1)$$

A decaying mismatch memory retains part of prior departures:

$$m_t = \lambda m_{t-1} + (1 - \lambda)e_t, \quad 0 \leq \lambda < 1. \quad (2)$$

The corrective update combines present mismatch and historical mismatch:

$$u_t = -K_p e_t - K_m m_t. \quad (3)$$

The system then evolves under its ordinary uncontrolled dynamics F_t , the correction, and disturbance η_t :

$$x_{t+1} = F_t(x_t) + u_t + \eta_t. \quad (4)$$

Over a horizon H , normalized trajectory error can be summarized by

$$D_H = \sqrt{\frac{1}{H} \sum_{t=1}^H \|x_t - r_t\|_2^2}. \quad (5)$$

Equation (2) gives the system more than a snapshot. It allows repeated small deviations in the same direction to accumulate into a stronger corrective context. That benefit is conditional. If K_m is too large, old mismatch can dominate current evidence, create overshoot, or destabilize the loop. Persistent correction therefore requires a balance between forgetting too quickly and reinforcing too strongly.

4. Bounded surrogate benchmark

4.1 Question and frozen design

The benchmark asks one narrow question: in a small transparent dynamical system, does retaining a decaying history of mismatch improve trajectory persistence beyond open-loop propagation and correction using only the current mismatch?

The target is a two-dimensional unit vector rotating at a fixed angular rate. The simulated system propagates the vector with biased angular velocity and additive Gaussian perturbation. It is tested under:

1. constant angular bias;
2. slowly varying angular bias;
3. a bias that reverses sign halfway through the run.

Each regime uses 200 fixed random seeds and 400 steps. Four controllers are evaluated: open loop, instantaneous proportional correction, replay-informed correction using Equation (2), and a separately disclosed unstable high-gain stress control. Mean trajectory RMSE is the primary endpoint. Tail RMSE, terminal error, and the fraction of seeds in which replay-informed correction beats instantaneous correction are secondary endpoints.

This benchmark is not an SVI reproduction. It does not use SVI weights, code, data, video metrics, or hardware. It also is not a test of a neural mechanism. It examines only the abstract control proposition defined in Equations (1)-(5).

4.2 Pilot disclosure and failure-control calibration

The first protocol designated a memory gain of 0.24 as an intended high-gain failure control. That value did not fail; it further improved tracking. The pilot protocol, script, and normal and optimized outputs are preserved. The result is important because it shows why a failure condition should not be invented after seeing a favorable ordinary-controller result.

A post-pilot diagnostic located an instability transition near a memory gain of 110. Version 2 left the primary controller, all disturbances, all seeds, and the decision endpoint unchanged. It changed only the separate stress-control gain to 110 so the manuscript could display an actual overcoupling boundary. That stress result is reported independently of the primary pass decision.

4.3 Results

Disturbance	Open-loop RMSE	Instantaneous RMSE	Replay-informed RMSE	Replay change vs instantaneous	Seeds favoring replay
Constant bias	1.20223	0.07359	0.06895	-6.30%	100%
Slowly varying bias	0.57299	0.04264	0.04004	-6.10%	100%
Sign reversal	0.67185	0.07165	0.06749	-5.80%	100%

Replay-informed correction reduced mean RMSE relative to open loop by 94.26%, 93.01%, and 89.95% in the three regimes. It also improved over instantaneous correction in every regime, including the sign reversal. Normal and optimized Python executions produced byte-identical result files.

The high-gain stress control was worse than ordinary replay-informed correction in all three regimes. Its mean RMSE was 2.91314 under constant bias, 0.52836 under slowly varying bias, and 0.35463 under sign reversal. The stress condition therefore illustrates the predicted failure boundary: mismatch memory can help at moderate coupling and destabilize the trajectory at excessive coupling.

4.4 What the result establishes

The benchmark establishes a mathematical possibility inside its frozen surrogate: a decaying record of prior mismatch can improve correction over present-error-only control, and excessive coupling to that record can be harmful. It does not establish that SVI's reported video results have been independently reproduced, that an Error Replay Memory is biologically implemented, or that the selected gains generalize beyond this synthetic system.

5. The biological problem

The brain also operates as a continuing system rather than a series of isolated frames. When the eyes move, a stable room must remain recognizable. When a hand reaches for a cup, the object's location, the body's

posture, the intended grip, and the consequence of movement must remain related across many updates. A later signal is meaningful partly because it belongs to an ongoing trajectory.

The biological question is therefore:

How could a distributed nervous system retain a context, detect a departure from that context, and use the history of departures to shape the next perception or action without requiring a separate inner observer?

Established neuroscience already supplies several components. Predictive-coding models assign predictions and residual errors to recurrent feedforward and feedback exchange [3]. Neural communication can depend on relative timing and coherence between participating populations [5]. Cross-frequency interactions can coordinate activity operating at different temporal scales [6,7]. Layer 6 corticothalamic projections and the thalamic reticular nucleus can modulate thalamocortical transmission, cortical population responses, and thalamic synchrony [8-12].

These findings do not by themselves join into a complete state-persistence operator. The task of the SAN proposal is to say what functional relationship those components might implement and how that relationship could be measured.

6. The proposed operation in plain language

Before introducing SAN terminology, consider the following sequence:

1. A receiving neural population is already in an ongoing state shaped by recent input, inhibition, membrane condition, expectations, and slower network rhythms.
2. A new signal arrives with a particular timing, phase relation, gain, and route.
3. The effect of the signal depends not only on the signal itself but on how it differs from the receiver's present and expected condition.
4. That difference changes which cells become recruited, which routes are suppressed, what information returns through recurrent loops, and what the system does next.
5. If a similar departure recurs, the system can retain a trace of it and adjust future sensitivity, routing, or action.
6. The next cycle tests whether the adjustment reduced the departure or merely displaced it.

The observer is not a hidden spectator inspecting this process. The distributed system registers the difference because the difference physically changes its subsequent state.

7. Phase Wave Differential

SAN calls the receiver-relative, consequential departure in Step 3 a **Phase Wave Differential (PWD)**. A useful operational form is

$$p_{ij}(t) = \text{wrap} \left[\phi_i(t - \tau_{ij}) - \hat{\phi}_{j|i}(t) \right], \quad (6)$$

where $\phi_i(t - \tau_{ij})$ is the arriving phase from sender i after route delay τ_{ij} , and $\hat{\phi}_{j|i}(t)$ is the phase relation expected by receiver j for that route and context. A richer receiver-relative update can include amplitude, gain, or state variables:

$$\mathbf{p}_{ij}(t) = \begin{bmatrix} \Delta\phi_{ij}(t) \\ \Delta a_{ij}(t) \\ \Delta g_{ij}(t) \\ \Delta s_j(t) \end{bmatrix}. \quad (7)$$

The proposal is not that every difference is an error. A deviation can be novelty, a sensory boundary, an action consequence, or a needed reset. Its informational role depends on the receiver and the trajectory. A locally useful PWD is a departure that measurably changes later recruitment, routing, plasticity, inference, or action.

This is why PWD is not simply another name for phase coding, traveling waves, or predictive coding:

Inherited term	What it ordinarily describes	Additional function proposed for PWD
Phase coding	Information represented by an event's relation to an oscillatory phase.	Defines a receiver-relative departure from an expected phase / state relation and follows its causal effect on the next update.
Traveling wave	A spatiotemporal pattern propagating across tissue.	Treats propagation as a carrier; the PWD is the consequential difference encountered by a particular receiving route.
Predictive coding	Recurrent exchange of predictions and residual errors.	Specifies an oscillatory, receiver-relative candidate variable through which a residual can be expressed and accumulated.
Communication through coherence	Selective communication enabled by compatible timing.	Includes both phase-compatible recruitment and informative departures that redirect or reset communication.

The distinction matters because a system that merely represents phase, propagates a wave, or computes a generic prediction error has not yet specified which receiver-relative difference is retained, how it changes a route, or how repeated differences become corrective context.

8. Two timescales and oscillatory state persistence

A fast event cannot define its own continuity. It requires a slower context against which “same,” “different,” “expected,” and “too far” can be judged. Let T_t denote a slowly changing contextual state and p_t a faster mismatch event:

$$T_{t+1} = \alpha T_t + (1 - \alpha)G(p_t, s_t), \quad \alpha \approx 1. \quad (8)$$

The slow state is not assumed to be one frequency band or one anatomical nucleus. It is a functional timescale that can be distributed across recurrent cortical, thalamic, hippocampal, cerebellar, bodily, and synaptic processes. Faster events update it; it in turn changes how future events are received.

The central two-timescale distinction is:

broad tonic coordination supplies a continuing reference, while faster phasic departures supply the differences that redraw, redirect, or update the trajectory.

Slow rhythms can coordinate windows of excitability while faster activity differentiates content within those windows. Cross-frequency coupling provides empirical examples of slower phase organizing faster amplitude [6,7]. Communication-through-coherence provides a framework in which compatible timing changes effective connectivity [5]. SAN adds the proposal that persistent, receiver-relative departures can be treated as a history-bearing correction signal rather than as isolated phase labels.

9. Mulling: correction that checks itself

One correction is not enough for a long-horizon system. A repaired state must be evaluated in the next cycle. The four-step process is:

1. retrieve the relevant state, expectation, or prior departure;
2. assess the present difference;
3. change the route, estimate, or action;
4. confirm whether the change preserved the intended trajectory.

After the operation is understood, SAN calls it **mulling**. Formally, the mismatch memory in Equation (2) is a minimal abstraction of its persistence, not a complete neural implementation.

Mulling stage	SVI engineering instance	Candidate biological instance
Retrieval	Sample error histories from Error Replay Memory.	Reactivate context through recurrent cortical, thalamic, hippocampal, or synaptic state.
Assessment	Compare error-injected latent state with clean target.	Compare arriving activity with receiver state and expected relation.
Correction	Learn error-recycled velocity.	Change gain, phase relation, routing, recruitment, prediction, or action.
Confirmation	Continue autoregressively and test long-horizon stability.	Use recurrent sensory and internal feedback to determine whether the departure was reduced or useful.

Correction costs resources. SVI spends training computation; biological systems spend metabolic energy and time. The thermodynamic statement here is therefore restrained: structured systems must perform work to prevent accumulated mismatch from erasing an organized trajectory.

10. Candidate corticothalamic implementation

The paper proposes a research map, not a completed circuit diagram:

$$\begin{aligned}
&\text{cortical commitment} \rightarrow \text{thalamic and TRN gating} \\
&\quad \rightarrow \text{Layer 6 feedback} \\
&\quad \rightarrow \text{cortical retuning} \\
&\quad \rightarrow \text{new sensory or internal evidence.}
\end{aligned} \tag{9}$$

Layer 5 pyramidal output is a candidate route by which cortical commitments influence subcortical and thalamic systems. The thalamus is a candidate hub for routing and modulating sensory, cortical, and recurrent context. The TRN provides inhibitory control over thalamic relay activity. Layer 6 corticothalamic projections can influence relay and TRN circuits and thereby alter the timing and effectiveness of thalamocortical transmission.

Primary studies constrain the map. McAlonan and colleagues found attentional modulation in primate TRN [8]. Crandall and colleagues described a frequency-dependent corticothalamic switch [9]. Guo and colleagues showed a corticothalamic circuit that dynamically switches auditory behavior between feature detection and discrimination [10]. Ibrahim and colleagues showed that Layer 6 corticothalamic feedback can gate population auditory thalamocortical transmission by changing thalamic synchrony [11]. Folkard and colleagues reported bidirectional control of cortical activity and thalamic firing mode by corticothalamic Layer 6 [12].

Together these results support timing-, synchrony-, and state-dependent corticothalamic control. They do not show that the thalamus is an SVI Error Replay Memory or that Equation (6) is already measured as a named biological variable. The experimental task is to determine whether receiver-relative departure and its history predict subsequent routing better than current activity alone.

11. Discriminating experiments

11.1 Neural state-persistence experiment

Record simultaneously from a cortical sender, a thalamic relay population, TRN, and a cortical receiver during a task that requires maintaining object or action identity across perturbations. Estimate route delay and expected phase / state relations from unperturbed trials. At perturbation time, compute the receiver-relative departure in Equations (6)-(7).

Compare models that predict the next receiver state from:

1. current activity alone;
2. current phase relation;
3. generic prediction error;
4. current receiver-relative PWD;
5. current PWD plus a decaying history of prior PWDs.

The state-persistence hypothesis gains support if Model 5 predicts corrective routing, recovery time, or behavioral continuity beyond Models 1-4 on held-out trials.

11.2 Causal perturbation

Perturb Layer 6 corticothalamic timing or TRN inhibition after a controlled departure. The hypothesis predicts a selective change in the use of mismatch history, not merely a global loss of arousal. Matched controls should preserve mean firing or total power while shifting the relative timing of the candidate route.

11.3 Failure boundary

The surrogate predicts a nonmonotonic relationship: too little persistence forgets repeated drift, while excessive persistence reinforces stale mismatch. A biological analogue would be supported by an intermediate regime that improves recovery and an overcoupled regime that produces overshoot, perseveration, instability, or failure to reset.

11.4 Falsifiers

The proposed operator should be demoted or revised if:

- current activity predicts correction as well as current-plus-history models across well-powered tasks;

- receiver-relative phase / state differences add no information beyond generic error or arousal;
- altering the candidate history trace changes neither recovery nor route selection;
- the proposed corticothalamic intervention produces only nonspecific impairment;
- the history term improves no held-out trajectory and has no identifiable failure boundary.

12. Implications for long-horizon AI

SVI suggests a general design principle:

Train systems on characteristic departures, retain a bounded correction state, estimate a direction of repair, and verify recovery across later steps.

For language, planning, robotics, and agentic systems, the principle separates five design responsibilities:

Responsibility	Function
Mission state Drift ledger	Preserve the task or intended trajectory across local steps. Record characteristic ways the system departs from that trajectory.
Error memory	Retain mismatch patterns for training or bounded runtime correction.
Correction direction	Estimate how to repair state, not merely whether state is wrong.
Confirmation loop	Test whether the repair survives subsequent interaction.

The high-gain result warns against unbounded self-correction. A system that treats every prior error as permanently authoritative can become less stable than one using no history. Error memory therefore needs decay, uncertainty, context separation, and a reset mechanism.

13. Source-version boundary

The mechanism analyzed here is fixed to arXiv:2510.09212v1, posted 10 October 2025 [1]. The official ICLR 2026 page confirms the title, authors, poster session, and oral presentation [2].

The implementation repository is a later and evolving source family. For this revision it is pinned to commit 63af54329f38b10ed1a91dc88bc67a765d0c4a18, whose GitHub record is dated 3 June 2026 [13]. At that snapshot the repository included SVI 1.0, SVI 2.0, SVI 2.0 Pro, Wan-related branches and workflows, community demonstrations, and EverAnimate-related routes. These later facts document engineering evolution; they do not retroactively enlarge the mechanism described in arXiv v1.

Evidence class	Authority	Conclusion permitted here
Fixed paper mechanism	arXiv v1	Describe ERFT, historical-error injection, error curation, replay memory, and error-recycled velocity.
Conference status	Official ICLR page	State the ICLR 2026 oral and poster status.
Later implementation family	Pinned repository commit	Describe later model, branch, and workflow existence with attribution.
Independent SVI reproduction	Absent from this package	No independent SVI metric or unlimited-duration result is claimed.
Bounded paper-level surrogate	Local protocol, code, and outputs	Report only the synthetic control result in Section 4.

14. Authorial intent and SAN genealogy

The paper’s conceptual ancestry predates the SVI comparison. It should not be reconstructed by assigning modern wording to old sources. Instead, each stage contributes a bounded atom:

Date	Source layer	Recoverable contribution
2017	Neural Lace / SAN public materials	Recurrent cortical-thalamic passes; distributed rendering; excitation and inhibition as information; tonic context and phasic difference.
2021-2022	SAN source-development corpus	Increasingly explicit treatment of recurrent oscillatory groups, mismatch, feedback, and distributed state.
June-August 2022	Publicly Git-fixed SAN notes	Mature PWD, tonic / phasic, oscillation-tomography, and receiver-relative update vocabulary.
2024-2025	SAN, thermodynamic, and research-program drafts	Structured mismatch dissipation, Layer 5/6-thalamus-TRN mapping, and correction across multiple timescales.
2025-2026	MASI Paper 03 development	Long-horizon persistence, slow mission state, fast task updates, SVI comparison, and mulling architecture joined into one paper.
2026	This successor	Version-scoped SVI audit, formal control abstraction, bounded benchmark, failure boundary, and discriminating experiments.

The genealogy preserves two different authorities. Public Git and publication records establish what was publicly fixed by a date. Private drafts and AI-chat exports can recover the development process but do not become public priority proof merely because they are locally dated. AI-generated suggestions are identified as development inputs, not silently reclassified as Micah-authored historical claims.

The intended thesis across these stages is affirmative: intelligence persists by retaining context, registering consequential departures, and transforming those departures into later correction. The purpose of the SVI comparison is to sharpen that mechanism with a concrete engineering case, not to replace SAN’s prior conceptual spine.

15. Claim register

Claim	Class	Current disposition
SVI addresses long-video drift using Error-Recycling Fine-Tuning.	Engineering fact	Supported by arXiv v1.
SVI identifies a clean-training / error-bearing-inference gap.	Engineering fact	Supported by arXiv v1.
SVI uses historical-error injection, bidirectional error curation, Error Replay Memory, and error-recycled velocity.	Engineering fact	Supported by arXiv v1.
The local surrogate shows a benefit from bounded mismatch memory.	Reproduced paper-level result	Supported only for the frozen two-dimensional benchmark.
Excessive memory coupling can destabilize the surrogate.	Reproduced failure condition	Supported for the disclosed high-gain stress control.

Claim	Class	Current disposition
Slow context and fast receiver-relative departures can form a state-persistence operator.	SAN theoretical claim	Formalized here; requires biological testing.
PWD differs functionally from phase coding, traveling waves, predictive coding, and coherence alone.	Construct distinction	Defined by receiver relativity, expected relation, causal update, and history.
Mulling is repeated retrieval, assessment, correction, and confirmation.	SAN working model	Operational definition; not asserted as a universal thermodynamic law.
Layer 6-thalamus-TRN dynamics are a candidate substrate for history-sensitive correction.	Neuroscience hypothesis	Motivated by primary corticothalamic studies; not established as the full operator.

16. Limitations

SVI’s author-reported benchmark results have not been independently reproduced in this package. The bounded surrogate is low-dimensional, synthetic, and supplied with the target trajectory. Its controller is designed rather than learned. Its result does not establish performance in video generation, language models, or neural tissue.

The neural analogy spans levels of description. Phase, firing, gain, synaptic state, thalamic routing, behavior, and subjective continuity are not interchangeable variables. Equations (6)-(8) specify candidates that must be tied to particular recordings and interventions. The circuit in Equation (9) is incomplete and should not be read as excluding hippocampal, basal-ganglia, cerebellar, neuromodulatory, glial, bodily, or other contributions.

The historical genealogy establishes development and public fixation where records permit. It does not by itself establish empirical truth, exclusive originality, or causal influence on later authors.

17. Conclusion

Long-horizon failure begins when a system treats each local output as sufficient. Continuity requires a reference state, detection of departure, memory of recurrent departure, a direction of correction, and a later test of whether the correction held.

SVI gives that problem an engineering center. It trains on self-generated historical error and learns recovery from corrupted latent state toward clean target. The local surrogate shows, in a deliberately small system, that a decaying mismatch history can improve trajectory persistence and that excessive coupling to memory can instead destabilize it.

SAN supplies a biological research program for the same functional burden. Slow recurrent context provides a candidate reference; faster receiver-relative departures provide candidate update signals; PWD names the consequential difference; mulling names repeated correction and confirmation; corticothalamic circuits provide one testable route. The next step is measurement: determine whether the history of receiver-relative departure predicts and causally shapes recovery beyond present activity, phase coding, generic prediction error, or arousal alone.

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